LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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Date: 31-08-2026

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1. TITLE

Design and Implementation of Styled Webpage with Canvas using HTML5 and CSS3

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2. AIM

To develop a styled, interactive HTML5 web page that implements advanced CSS3 styling, flexbox layouts, gradient backgrounds, hover effects, and uses the HTML5 Canvas API to render basic 2D graphics.

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3. OBJECTIVE

The objectives of this experiment are:

- To master CSS3 styling including gradients, borders, shadows, and rounded corners.
- To utilize CSS Flexbox for creating responsive, structured dashboard layouts.
- To implement interactive hover effects and transitions on web elements.
- To draw shapes, lines, and text programmatically using the HTML5 Canvas 2D context.

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4. THEORY

CSS3 extends cascading style sheets with features like rounded corners, shadows, gradients, transitions, and flexbox layouts. Flexbox (Flexible Box Layout) makes it easier to design flexible responsive layout structures without using floats or positioning.

The HTML5 `<canvas>` element is used to draw graphics on a web page via JavaScript. The canvas API provides a 2D context allowing developers to draw shapes, lines, arcs, and text, rendering them pixel-by-pixel dynamically.

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5. ALGORITHM / PROCEDURE

1. Create an HTML5 document with a navigation bar and dashboard section.
2. Apply a linear gradient background to the body using CSS.
3. Style the navigation and feature cards using flexbox, padding, borders, and box-shadows.
4. Add CSS transitions to button elements for smooth hover scaling effects.
5. Include a `<canvas>` element with defined width and height.
6. Write a JavaScript block to obtain the canvas 2D context.
7. Use context methods (fillRect, arc, lineTo, fillText) to draw a rectangle, a circle, a line, and a name on the canvas.
8. Open in web browser and verify styling and canvas rendering.

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6. PROGRAM

Source File: task2.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Task 2 - Styled Webpage</title>
    <style>
        /* CSS Selectors, Colors, Fonts, and Gradient Background */
        body {
            font-family: Arial, sans-serif;
            background: linear-gradient(135deg, #c78686, #f5ff9a);
            margin: 0;
            padding: 20px;
            min-height: 100vh;
        }

        /* Header with Text Shadow */
        header {
            text-align: center;
            margin-bottom: 20px;
        }

        header h1 {
            font-family: 'Courier New', Courier, monospace;
            font-size: 32px;
            color: #181818;
            text-shadow: 2px 2px 8px rgba(0, 0, 0, 0.4);
        }

        /* Navigation Bar */
        #main-nav {
            background-color: #f5ff9a;
            padding: 15px;
            border: 4px solid black;
            border-radius: 8px;
            color: rgb(24, 24, 24);
            text-align: center;
            box-shadow: 0px 4px 6px rgba(36, 41, 28, 0.3);
            margin-bottom: 30px;
        }

        #main-nav a {
            color: #2c3e50;
            text-decoration: none;
            font-weight: bold;
            margin: 0 15px;
            font-size: 16px;
        }

        #main-nav a:hover {
            color: #ff007f;
        }
```

```
/* Canvas Section Styling with Borders and Shadows */
.canvas-container {
    text-align: center;
    background-color: rgba(255, 255, 255, 0.85);
    padding: 20px;
    border-radius: 12px;
    border: 3px solid #181818;
    margin-bottom: 30px;
    box-shadow: 0px 4px 10px rgba(0,0,0,0.2);
}

canvas {
    border: 3px solid black;
    border-radius: 8px;
    background-color: #ffffff;
}

/* Three Content Cards Container */
.dashboard {
    display: flex;
    justify-content: space-around;
    align-items: center;
    gap: 20px;
    margin-bottom: 40px;
    flex-wrap: wrap;
}

/* Feature Cards with Borders, Rounded Corners, Padding, and Box Shadow */
.feature-card {
    background-color: rgb(255, 98, 106);
    border: 2px solid #92e0da;
    border-radius: 12px;
    padding: 22px;
    margin: 8px;
    box-shadow: 0px 4px 10px #32413f;
    flex: 1;
    min-width: 220px;
    text-align: center;
    transition: transform 0.3s ease;
}

/* Hover Effect for Cards */
.feature-card:hover {
    transform: translateY(-5px);
}

.feature-card h3 {
    font-family: 'Verdana', sans-serif;
    font-size: 20px;
    color: #fff;
}

.feature-card p {
    font-size: 14px;
```

```
            color: #fce4ec;
        }

        /* Button Styling with Hover Effects */
        .btn-p {
            background-color: bisque;
            color: rgb(37, 19, 104);
            padding: 10px 16px;
            border: none;
            border-radius: 5px;
            cursor: pointer;
            box-shadow: 0px 4px 8px rgba(7, 4, 26, 0.3);
            font-weight: bold;
            transition: background-color 0.3s ease, transform 0.2s ease;
        }

        .btn-p:hover {
            background-color: #ffeb3b;
            transform: scale(1.05);
        }

        /* Footer Component */
        footer {
            text-align: center;
            padding: 15px;
            background-color: #181818;
            color: #f5ff9a;
            border-radius: 8px;
            font-size: 14px;
        }
    </style>
</head>
<body>

    <!-- Header Component -->
    <header>
        <h1>My Styled Webpage - Task 2</h1>
    </header>

    <!-- Navigation Bar Component -->
    <nav id="main-nav">
        <strong>IDEAS TO IMPLEMENT</strong><br><br>
        <a href="#">Home</a>
        <a href="#">Graphics</a>
        <a href="#">Dashboard</a>
    </nav>

    <!-- Part A: HTML5 Canvas Element -->
    <div class="canvas-container">
        <h3>HTML5 Graphics Canvas</h3>
        <canvas id="taskCanvas" width="500" height="200"></canvas>
    </div>

    <!-- Part B: Three Content Cards Component -->
```

```
    <div class="dashboard">
        <div class="feature-card">
            <h3>AI Innovation</h3>
            <p>Create next-gen AI model ideas and logic frameworks.</p>
            <button class="btn-p">Open</button>
        </div>

        <div class="feature-card">
            <h3>ai gen</h3>
            <p>Design combat mechanics and interactive levels.</p>
            <button class="btn-p">Open</button>
        </div>

        <div class="feature-card">
            <h3>Projects</h3>
            <p>Build and deploy high-performance web solutions.</p>
            <button class="btn-p">Open</button>
        </div>
    </div>

    <!-- Footer Component -->
    <footer>
        <p>&copy; 2026 Developed by A.Rohit | ANITS Web Development Task</p>
    </footer>

    <!-- Canvas Drawing Script -->
    <script>
        const canvas = document.getElementById('taskCanvas');
        const ctx = canvas.getContext('2d');

        // 1. Draw a Rectangle
        ctx.fillStyle = '#ff5722';
        ctx.fillRect(30, 30, 100, 60);

        // 2. Draw a Circle
        ctx.beginPath();
        ctx.arc(200, 60, 35, 0, 2 * Math.PI);
        ctx.fillStyle = '#4caf50';
        ctx.fill();
        ctx.lineWidth = 2;
        ctx.strokeStyle = '#2e7d32';
        ctx.stroke();

        // 3. Draw a Line
        ctx.beginPath();
        ctx.moveTo(270, 30);
        ctx.lineTo(370, 90);
        ctx.lineWidth = 4;
        ctx.strokeStyle = '#3f51b5';
        ctx.stroke();

        // 4. Display Name as Text
        ctx.font = 'bold 22px Arial';
        ctx.fillStyle = '#181818';
```

```
        ctx.fillText('A.Rohit', 150, 160);
    </script>

</body>
</html>
```

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7. CODE EXPLANATION

Code / Syntax	Explanation
linear-gradient()	CSS function to create a background image of linear color transitions.
box-shadow	Applies drop shadow effects to element bounding boxes.
display: flex;	Enables flexbox layout container for easy alignment of child elements.
canvas.getContext('2d')	Returns a two-dimensional rendering context for the canvas element.
ctx.fillRect()	Draws a filled rectangle on the canvas.
ctx.arc()	Adds an arc/curve (circle) to the current sub-path.
ctx.fillText()	Draws filled text on the canvas at specified coordinates.

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8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	CSS Styling	Verify page background and borders	Gradients and shadows render correctly	Pass
TC-02	Hover Effects	Hover mouse over buttons/cards	Elements smoothly scale up on hover	Pass
TC-03	Canvas Context	Check canvas drawing execution	JavaScript correctly loads 2D context	Pass
TC-04	Graphics Draw	Verify drawn shapes and text	Rectangle, circle, line, and name visible	Pass

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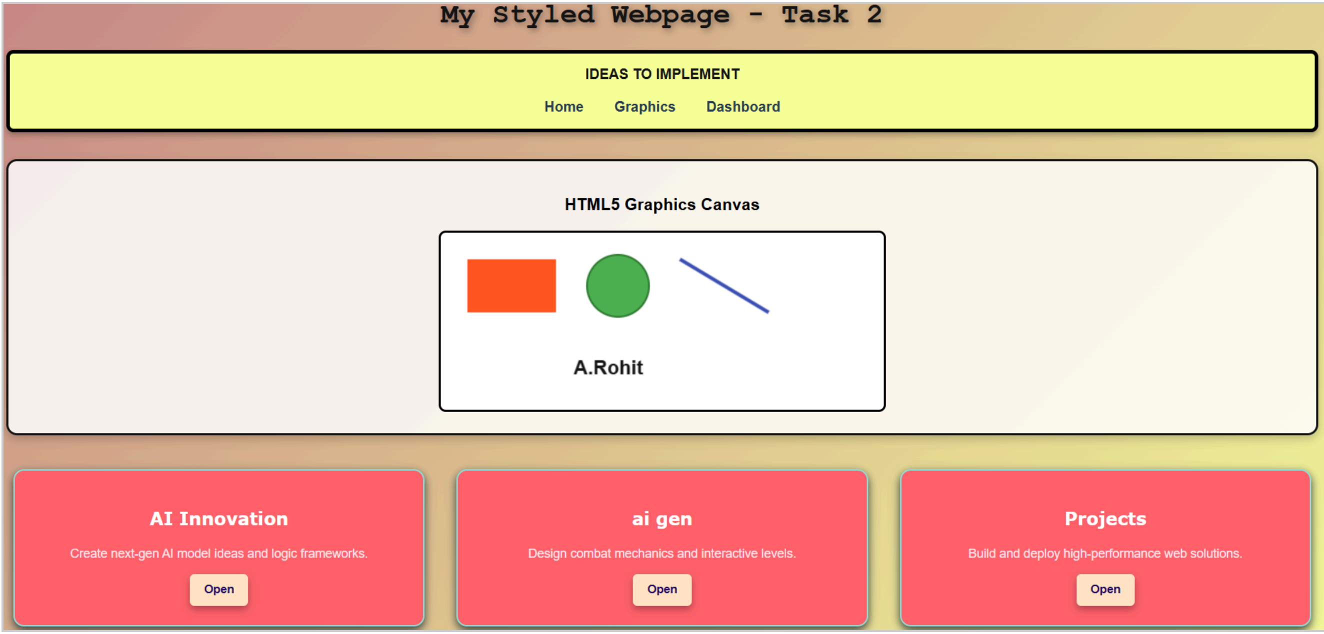
9. OUTPUT

Output 1



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10. RESULT

Thus, the styled webpage with CSS3 layouts and HTML5 Canvas graphics was successfully implemented and verified across all test cases.